

Description of the *MilkyBase* database

Pacza T. et. al., 2022: MilkyBase, a database of human milk composition as a function of maternal-, infant- and measurement conditions. Nature Scientific Data

Introduction

The ***MilkyBase*** database hosts records on the biochemical composition of human milk as a response to specified factors of the mother and infant and some characteristics how and where the data were generated. This manual presents the ontology of the database and provides a technical description for its records.

Rationale for a common data format

Food composition data are presented in various formats. It is paramount to standardize these formats, to make the databases compatible and utilize them to their full potentials. A uniform structure facilitates comparisons within and between datasets and allows rapid search in the database by pre-determined key parameters.

Overview of the database ontology

The database is basically a system of connected sheets in a single Microsoft Excel workbook. Throughout the following Description, words in bold, with capitalised initials refer to **Sheets**, and italic *Courier* type refer to *fields*.

MilkyBase is a flat database. Its **Master** sheet consists of records identified by unique keys. The main fields can be grouped into Admin details (specifications how the data were acquired), Conditions and Components (playing the role of explanatory and response variables). The value set of the respective possible entries has a tree-structure. For example, the entries in Components field can be single chemicals (leaf, like “VitC”), or a group of chemicals (a branch like Vitamins). See **Figure 1**.

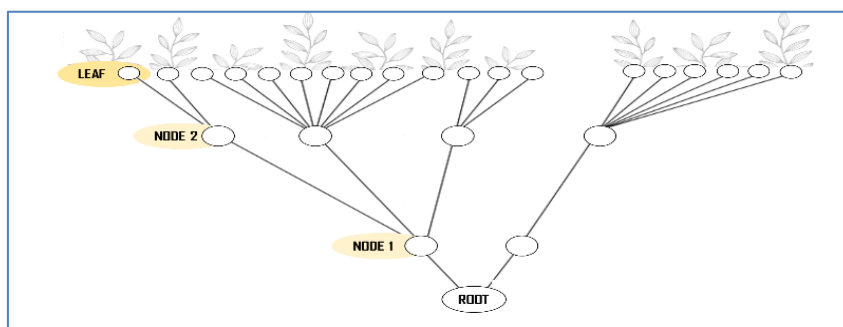


Figure 1. A tree-structured value set

The names of the fields of the **Master** sheet are listed in the **Field** sheet. The names of the sheets containing the respective value sets are also listed there.

Field sheet

Summary: The **Field** sheet contains definitions and descriptions for the fields of the **Master** and **Source** sheets of the database.

	A	B	C	D	E	F	G	H
1	Field	Alternative	Group	Description	ValueSet	Unit	TYPE	Comment
2	Key		Field	Unique record ID in the Master sheet			Alphanumeric	Unique one word (can be an integer, too); possibly
3	Food		Field	Food name	HumanMilk; BovineMilk; InfantFormula		Alphanumeric	Likely to be extended, in which case ValueSet is bet
4	Source		Field	Source of information	_Source		Alphanumeric	Also behaves like a Master sheet
5	Region		Field	Geographic region of cohort	_Region		Alphanumeric	Tree-structured definition sheet
6	CohortSize		Field	Cohort size	[1, 99999]		Numeric	
7	MeasMethod	MeasurementMethod	Field	Method of collecting data	MeasMethod		Alphanumeric	Tree-structured definition sheet

Figure 2. The **Field** sheet

The first column of the **Field** sheet contains the field names used in the **Master** sheet of the database. The heading (column names) is as follows.

Field | **Alternative** | **Group** | **Description** | **ValueSet** | **Unit** | **TYPE** | **Comment**

- *Field*: The name of the field, for which the description is given.
- *Alternative*: Alternative name(s) for the *Field*.
- *Group*: Hierarchical (tree structured) grouping information.
- *Description*: Brief description for the field in question.
- *ValueSet*: Describes the range/type of values allowed for the field.
- *Unit*: The unit if the field value is a quantity.
- *Type*: Defines the field type.

When checking the syntax of the entries in the **Master** sheet, their field names are looked up in the **Field** sheet. Therefore, if a new field is needed in the **Master** sheet, that should be defined in the **Field** sheet first (purple tab).

Master sheet

Summary: This is the main sheet. The columns represent fields, their headings are the field names. The allowed values in a cell are defined in definition sheets, each with the same name (see its tab) as the field name in question (**Figure 3**). The possible field names are listed in the **Field** sheet.

Key	Food	Source	Region	CohortSize	MeasMethod	Condition	Component
HM-TP-Bri-86-01	HumanMilk	Britton_86	AZ	70	Lowry; CC-UV; SDS	GestAge(week)=[25,35] MilkStage(4	Prot(g/L)=!Britton_Prot AspA_mol(m
HM-TP-Bri-86-02	HumanMilk	Britton_86	AZ	38	Lowry; CC-UV; SDS	GestAge(week)=[38,42] MilkStage(4	Prot(g/L)=!Britton_Prot AspA_mol(m
HM-MM-But-84a-01	HumanMilk	Butte_84a	TX	8	ABCM; KM; Colorim	GestAge(week)=33.9±2.3 Age_m(y	Fat(g/L)=!Butte84aP_Fat Ca(g/L)=!But
HM-MM-But-84a-02	HumanMilk	Butte_84a	TX	13	ABCM; KM; Colorim	GestAge(week)=39.2±1.4 Age_m(y	Fat(g/L)=!Butte84aF_Fat Ca(g/L)=!But
HM-TP-But-84b-01	HumanMilk	Butte_84b	TX	45	ABCM; KM; RGP; GL	Age_m(year)=18.0±3.1 Weight_c(g	Fat(g/L)=!Butte84b_Fat N(g/L)=!Butte
HM-MM-But-90-01	HumanMilk	Butte_90	TX	10	ABCM; KM; modify	Weight_c(g)=33.45±335 GestAge(w	Fat(g/L)=33.53±7.75 N(g/L)=2.1±0.46
HM-MM-But-90-02	HumanMilk	Butte_90	TX	10	ABCM; KM; modify	Weight_c(g)=34.9±336 GestAge(w	Fat(g/L)=28.04±8.43 N(g/L)=1.73±0.14
HM-TP-Cam-09-01	HumanMilk	Campbell_09	Halifax	22	KM; RGP; FAOM; CR	MilkStage(day)=[0,14] Domperidon	Fat(g/L)=!CampbellDomp_Fat Prot(g/

Figure 3. A section of the **Master** sheet. The possible values of a field (whose name is in the heading of the respective column) are stored in a separate, so-called definition-sheet, with the same name. The arrows point to the tabs of the respective definition sheets.

The **Master** sheet has these headings:

Key | Food | Source | Region | CohortSize(cap) | MeasMethod | Condition | Component | Comment

- **Key:** A unique identifier for each record in the **Master** sheet.

For example, our current practice is to build the key from 5 parts separated by hyphens (-).

AA-BB-Ccc-xx-yy, where

- **AA** – Code for the Food (HM for HumanMilk; with the potential in mind that, sometime in the future, the composition of bovine, plant-based, etc milk may also be recorded in this database,)
- **BB** - Initials of the person who inputted the record (e.g., JS for John Smith)
- **Ccc** - First three letters of the surname of the first author of the paper, whose data are recorded (e.g., Dan for Daniels)
- **xx** - Year of the publication of the source (e.g., 04 represents 2004)
- **yy** – An ID number within the source (e.g., 01, 02...) making sure that the key is unique. (e.g., HM-JS-Dan-04-01, HM-JS-Dan-04-02, ...)
- **Food:** Name or category of the studied food.
Currently MilkyBase only accepts HumanMilk keyword for this; later we may extend the database to Infant Formula, Bovine Milk... etc; see the **Field** sheet.
- **Source:** The source of the information stored by the record. It is an abbreviation defined in the **Source** sheet.
- **Region:** Geographic region of the cohort. The (tree-structured) value set is defined in the **Region** sheet.

- *CohortSize (cap)*: Size of the cohort measured in *capita*. Typically, this is also the sample size when averages are taken for the components. It must be a positive integer; see the **Field** sheet.
- *MeasMethod*: Measurement details are available here.
- *Condition (equivalent to an explanatory vector-variable)*: A vector-field, where the entries, separated by vertical bars (|), are various conditions. Therefore, the whole vector is collected in a single cell. Basically, the vertical line behaves like a separator between fields, for which the valid entries are listed in the **Condition** sheet, like variable names (e.g., the child's weight or certain lifestyle-features of the mother).
- *Component*: Like *Condition*, this is also a vector-field. Its entries are quantified (g / litre milk by default) biochemical components of the food (e.g., the vitamin B1, or thiamine) or groups of compounds (e.g., the group of the vitamin B complex). The value set is stored in the **Component** sheet.

Definition sheets

Summary: The function of all the other sheets is to show further details on the **Master** sheet entries.

The name of each definition sheet (see its tab) is also displayed in the cell A1 of the sheet. In this column, alphanumeric strings are stored, which behave like mathematical variables. Underscore (_) or hyphen (-) should be used for separators, instead of blanks.

In the second column *Alternative* names can also be added, with more interpretable longer names. The third field is called *Group*; the name of the group can be stored here, which the entry belongs to. As has been said, this is the computational realization of the tree-structure, where the root, nodes and leaves represent the studied food, the groups of its components and the components themselves, respectively (**Figure 4**).

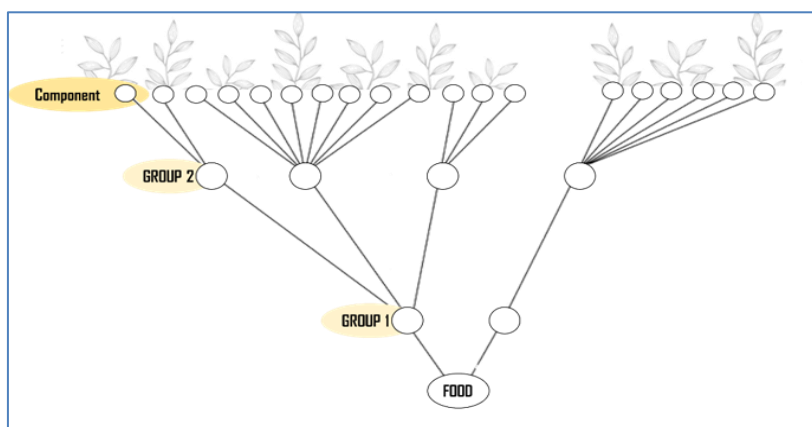


Figure 4. Tree structure of the value set of the **Components** sheet

The fourth column contains free text for *Description*.

The fifth column is the *ValueSet*: this defines the categories (alphanumeric type) and intervals (numeric type), from which a value is inputted in the **Master** sheet. **Figure 5** shows all the discussed fields of a definition sheet.

	A	B	C	D	E	F	G
1	Component	Alternative	Group	Description	ValueSet	Unit	Type
41	AlaA	Alanine	TAA	Alanine is an amino acid that is used to make protein	[0, 1e10]	g/L	Numeric

Figure 5. Headings of a definition sheet

Values

Allowed values can be subdivided into these types; *alphanumeric*, *numeric*, *Boolean*, *FreeText* or *Group*.

- Alphanumeric value

A word or a list of words consisting of standard letters and numbers (e.g. “allergy; asthma; IBD; CVD; HIV”). If it is a list of such values, then the separator is semicolon (;)

	A	B	C	D	E	F	
1	Key	Food	Source	Region	Cohort	MeasMethod	Condition
125	HM-TP-Eli-11-01	HumanMi	Elisia_11	Canada	10	CREAM; GC-FID; HPLC	MilkStage(day)=14 MilkTyp
126	HM-MM-Elw-19-01	HumanMi	Elwakiel_19	Hohhot	7	MS	MilkStage(day)=[1,7]
127	HM-MM-Elw-19-02	HumanMi	Elwakiel_19	Netherlands	4	MS	MilkStage(day)=[1,7]

Figure 6. Alphanumeric values in the **Master** sheet

The first four fields in the **Master** sheet are alphanumeric fields (**Figure 6**).

When a *ValueSet* is a list (set) of allowed items, or a long description, then an initial underscore refers to a Table (sheet) of the same name in the **Field** sheet. A value like “_Region” means that the acceptable values are listed in the sheet called “**Region**”.

- List: Series of alphanumeric values (items) separated by semicolons.

E.g.: The *ValueSet* for the condition HealthIssue_m is “allergy; asthma; IBD; CVD; HIV” and, for the LifeStyle_m, it is: “smoker; physAct”.

- The *Unit* field for Alphanumeric values must be empty. See **Figure 7**.

	A	B	C	D	E	F	G
1	Condition ▾	Alternative ▾	Group ▾	Description	ValueSet ▾	Unit ▾	TYPE ▾
9	HealthIssue_m	HealthIssue_mother	COND_m	Issue with the Mother's health	{allergy; asthma; IBD; CVD; HIV}		AlphaNumeric
10	LifeStyle_m	LifeStyle_mother	COND_m	Mother's lifestyle features	{smoker; physAct}		AlphaNumeric
11	VitDefic_m	Vitamin_deficiency	COND_m	Deficiency of vitamins for mother	{A; B1; B2; B3; B5; B6; B12; C; D; E; K}		AlphaNumeric

Figure 7. Alphanumeric values in the **Condition** sheet

- Numeric value

This can be integer (here 9 is used as a generic symbol), fixed point format (like 999.99), exponential format (like -9.99e9) and interval (like [9e8, 9.9e9]). The Hashtag (#) is a special character, indicating a placeholder for an unknown number. So [10, #] indicates a number greater than 10. In the **Condition** and **Component** sheets, the values can be supplied with error margins like 5 ± 1 or $7@[3,9]$, the latter one meaning that either (i) the real value is between 3 and 9; or (ii) 7 is a (not necessarily arithmetical) centre with [3,9] as quantiles; or (iii) if 7 is an estimation then its 95% confidence interval is [3,9]).

Scientific notation is used when a number is less than 0.001.

In the **Condition** and the **Component** sheets, a special, dual format is also allowed. It consists of two parts, separated by a semicolon. The first part refers to **raw (measured)** data, possibly with its **scatter** (either its standard deviation, or Inter-Quantile range or a [Min, Max] interval); the second refers to an **estimation** and the **confidence in this estimation** (either standard error or 95% confidence interval).

Therefore, the first part is **descriptive** (typically the average of observations and a quantification of the scatter around it), while the second part is **predictive (or estimating)** with the standard error or confidence interval of the prediction/estimation.

E.g., see below the interpretation of $X \pm Y$; $X@[x,y]$

- In the first part, **X** can be any centroid of the data (Mean, Median, Mode, Geometric mean, etc.), which is about the raw observations. If **X** is in the second part, then it is an estimation / prediction (of the mean, generally).
- **Y** is the Standard Deviation of the raw data (if it is in the first part); or the standard error of the estimate (if it is in the second part)
- If an **[x, y]** interval is in the first part, then it indicates either Range [Min, Max], or Interquartile Range (IQR, equalling to the interval defined by the 75th and 25th percentiles), or any other Quantile Intervals (most likely 95% QI). If it is in the second part, then it denotes a Confidence Interval (normally 95% CI).

In the other fields, the content of a numeric field can be a number x; or an interval [x, y], e.g. [9e8, 9e9]. See an example of published data^{1,2} below. (**Figure 8, Figure 9**)

Table 2. Concentrations (median, mean, range) of trace elements (µg/kg) for human milk and serum for healthy adults			
Element	Colostrum 1-3 d post partum (n=13)	Transitory milk 4-17 d (n=18)	Mature milk 42-60 d (M1) (n=8)
Ba 014 ^a	14	3.0	2.1
	19	4.1	2.8
	0.9-47	1.0-12 0.47-2.4***	1.0-5.4
Be 0.05 ^a	0.4	<0.05	<0.05
	0.4	0.1	0.1
	<0.05-0.6 0.04-0.3**	<0.05-0.7	<0.05-0.8

Figure 8. $Ba(g/L) = 1.4E-5 @ [9.00E-7, 4.70E-5]$ | $Be(g/L) = 4.00E-7 @ [0, 6.00E-7]$ ¹

TABLE 2 <i>Micronutrient profile of human colostrum</i>			
Micronutrient in µmol/l	Number (%)	Mean	SD
Vitamin E			
6.00–18.80	37 (35)		
18.80–30.00	52 (50)	21.34	8.47
30.00 and above	16 (15)		
Vitamin C			
70.00–116.00	29 (28)		
116.00–162.00	35 (33)	148.92	43.64
162 and above	41 (39)		

Figure 9. $VitE_mol(mol/L) = 2.134E-5 \pm 8.47E-5$ | $VitC_mol(mol/L) = 1.4892E-4 \pm 4.4364E-5$ ²

- Boolean value

As in MS Excel, two values are allowed for Boolean variables: *true* and *false*.

Source sheet

Summary: This sheet contains information about the source of the collected data – usually papers. (Figure 10)

1	Source	Title	SourceDOI	PublDate(year)	InputDate(year)	InputBy	Comment/Summary
80	Macias_01	Changes in the Concentration of Carotenoids	doi.org/10.1159/000046711	2001	2021	UD	
81	Maly_18	Preterm human milk macronutrient concentrations	doi.org/10.1136/archdischild-2016-31257	2018	2020	UD	Study evaluated the amount of mac
82	Martysiak-Zurowski	Concentrations of alpha- and gamma-tocopherol	doi.org/10.1111/j.1740-8709.2012.00400.x	2012	2021	UD	
83	Mateo-Vivas_15	Analysis of Free Nucleotide Monophosphates	doi.org/10.1016/j.foodchem.2014.11.051	2015	2021	UD	
84	Michaelsen_90	Variation in Macronutrients in Human Breast Milk	doi.org/10.1097/00005176-199008000-00000	1990	2021	UD	
85	Montagne_99	Immunological and nutritional composition of human milk	doi.org/10.1097/00005176-199907000-00000	1999	2020	UD	Immunologic and nutritional protein
86	Morrow_04	Human milk oligosaccharides are associated with infant growth	doi.org/10.1016/j.jpeds.2004.04.054	2004	2020	UD	The aim was to determine the assoc
87	Motil_97	Human Milk Protein Does Not Limit Growth	doi.org/10.1097/00005176-199701000-00000	1997	2021	UD	
88	Nommensen_91	Determinants of energy protein lipid and carbohydrate content of human milk	doi.org/10.1093/ajcn/53.2.457	1991	2020	UD	Macronutrients composition of Hun
89	Olafsdottir_06	Polyunsaturated fatty acids in the diet and in human milk	doi.org/10.1159/000091685	2006	2022	UD	
90	Oliveira_19	Development of a human milk concentration scale	doi.org/10.1371/journal.pone.0210999	2019	2022	UD	Constituting a lyophilized human m
91	Orhon_08	The influence of maternal smoking on milk composition	doi.org/10.1007/s00431-008-0873-0	2008	2021	UD	

Figure 10. The Source sheet

The Source sheet has heading (field names) as below:

Source | Title | SourceDOI | PublDate(year) | InputDate(year) | InputBy | Comment/Summary

- *Source*: ID of the source record linked to the Source column of the Master sheet.
- *Title*: Title of the paper, study, report, or website
- *SourceDOI*: DOI of the web document
- *PublDate(year)*: Publication year
- *InputDate(year)*: Year when the record was inputted in the database
- *InputBy*: ID of the responsible for the record. The value set is in the field named InputBy.

Region sheet

Summary: The Region sheet details the possible values that the Region field of the Master sheet can take up. It has the headings (field names):

Region | Alternative | Group | Description | Comment

- *Region*: Region (country, province, city, etc) where the cohort came from (e.g., USA). It links the entries in the Master sheet to their definition in the Region sheet.
- *Alternative*: Other alternative names for the region (i.e.: an alternative name for the UK is “United_Kingdom”)
- *Group*: Group, in the tree structure, where the region belongs to. For example: USA belongs to the North America - “NorthAm” - which belongs to the root “World”
- *Description*: Short description of the region.
- *Comment*: Free text on any additional information.

MeasMethod sheet

Summary: The MeasMethod sheet details the measurement used to generate the response data of the record.

	A	B	C	D	E
1	MeasMethod	Alternative	Group	Description	Comment
2	AAA		IEC-Ninhydrin	JLC-500V amino acid analyzer	https://doi.org/10.1016/j
3	AAS		AS	Atomic absorption spectrometry (AAS) is a very sensitive method of elemental	belongs to Spectrometry.
4	ABCM		Calorimetry	Adiabatic Bomb Calorimeter	https://www.researchga
5	ABTS		Colorimetry	2-20-azinobis 3-ethylbenzothiazoline-6-sulfonic acid radical cation assay	https://doi.org/10.1080/
6	AEM		Chronoamperometry	Amperometric enzymatic method to analyze lactose content	https://doi.org/10.1016/j
7	Amperometry		Electrochem	Amperometry is an electroanalytical technique that involves the application of a constant reducing or ox	

Figure 11. The MeasMethod sheet

The MeasMethod sheet (Figure 11) has the heading (field names):

MeasMethod | Alternative | Group | Description | comment

- *MeasMethod*: The Method that was used to measure the components (e.g., Capillary Gas Chromatography, “CGC”). It links the entries in the **Master** sheet to their definition in the **MeasMethod** sheet.
- *Alternative*: Other alternative names for the method (e.g.: alternative name for the CGC: “Capillary_Gas_Chromatography”)
- *Group*: The name of the group in the tree structure where the method belongs to. For example: CGC belongs to the methods of Gas Chromatography, the “GC” group, which belongs to the root “MeasMethod” (Figure 12).

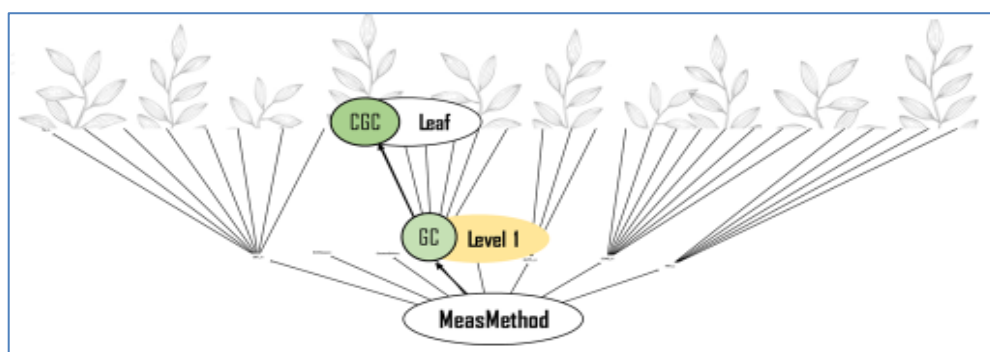


Figure 12. Hierarchical organisation of the Capillary Gas Chromatography method in the MeasMethod sheet.

- *Description*: Short description of the method.
- *Comment*: Free text about any important information such as DOI of the web document describing the method.

Condition sheet

Summary: Conditions are like explanatory variables in maths. They are factors affecting the response, i.e., the milk composition. They can be related to the mother (e.g., gestational age, weight, lifestyle, supplementation), to the infant at birth (e.g., weight, health issue, length) and to the milk sample (e.g., pasteurization, stage of lactation, storage temperature). See **Figure 13**.

	A	B	C	D	E	F	G	H
1	Condition	Alternative	Group	Description	ValueSet	Unit	TYPE	Comment
2	COND_m	Condition_mother	Condition	Various info on the condition of Mother				
3	GestAge	Gestational_Age	COND_m	Gestational age refers to the 'age' of the neon	[15, 50]	week	Numeric	
4	Age_m	AgeAtBirth_mother	COND_m	Mother's (or cohort's mean-) age at child birth	[10, 60]	year	Numeric	
5	BMI_m	BMI_mother	COND_m	Mother's BMI. Default: Pre-pregnancy	[5, 50]	kg/m2	Numeric	Formula: BMI=kg/m². Can
6	Height_m	Height_mother	COND_m	Mother's height	[50, 250]	cm	Numeric	Can be dynamic.
7	Weight_m	PrePregnancyWeight_mother	COND_m	Weight of Mother. Default: pre-pregnancy	[0, 200]	kg	Numeric	Can be dynamic.

Figure 13 . The *Condition* sheet

- The **Condition** sheet has the heading (field names) as below:

**Condition | Alternative | Group | Description | ValueSet | Unit |
TYPE | Comment**

- *Condition*: Condition name. It links the entries in the *Condition* field of the **Master** sheet to the **Condition** sheet.
- *Alternative*: alternative name(s).
- *Group*: the name of the group which the condition belongs, following the usual tree structure. For example, the weight of the child at birth, “weight_c” belongs to the child’s information, “INFO_c”, which in turn belongs to the root condition (**Figure 14**).

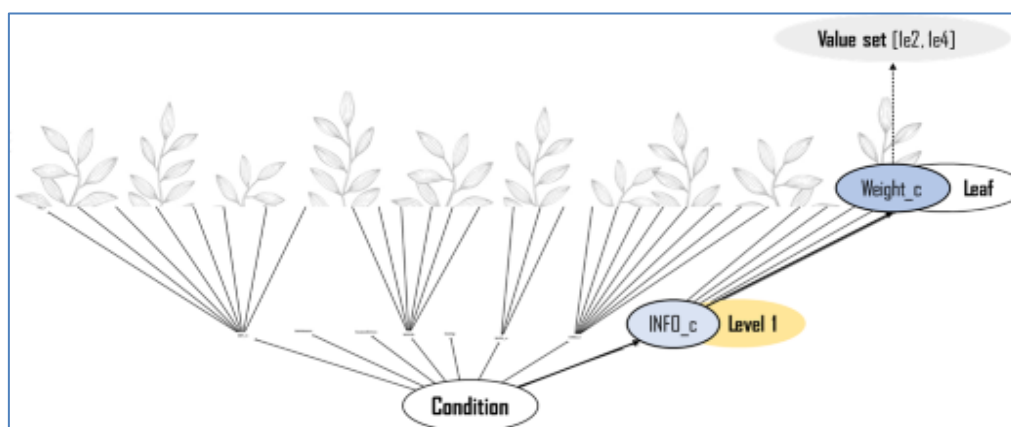


Figure 14. Hierarchical organisation of the value set of the child’s weight among the conditions.

- *Description*: Brief description of the condition.

- *ValueSet*: Values for the condition (e.g., “{allergy; asthma; IBD; CVD; HIV}”, for category values).
- *Unit*: Unit used for the condition. Allowed units for numeric values are in the **Unit** sheet. There is no unit for alphanumeric values.
- *TYPE*: Type of the value in the ValueSet (e.g., Alphanumeric, numeric, Boolean)
- *Comment*: free text about any issue of the condition in question.

Component sheet

Summary: Food components are like response variables depending on the conditions. Its value set is stored in the **Component** sheet.

	A	B	C	D	E	F	G	H	I	J
1	Component	Alternative	Group	Description	ValueSet	Unit	Type	MolWeight	ChemID	Comment
2	25OHD2		VitD	25-Hydroxyvitamin-D2. A metabolite of vitamin	[0, 1.e-3]	g/L	Numeric	MolWeight(g/mol)=412.6	InchiKey=KJIIUAXZGLUND-ICCVIKNSA-N	
3	25OHD3	Calcifediol	VitD	25-Hydroxyvitamin-D3. A metabolite of vitamin	[0, 1.e-3]	g/L	Numeric	MolWeight(g/mol)=400.6	InchiKey=JWUBBDSIWLEOM-DTOXIADCSA-N	
4	2FL		FS-OS	2-Fucosylated lactose	[0, 100]	g/L	Numeric	MolWeight(g/mol)=488.4	InchiKey=SNFSYLYCDAVZGP-ZTFPOLCTSA-N	
5	2FL_mol				[0, 0.1]	mol/L	Numeric			
6	3FL		neutOS	3-Fucosyllactose	[0, 100]	g/L	Numeric	MolWeight(g/mol)=488.4	InchiKey=WJPIUUDKRHCAEL-YVEAQFMBSA-N	
7	3FL_mol				[0, 0.1]	mol/L	Numeric			
8	3SL		SL-OS	3-Sialyl Lactose	[0, 1]	g/L	Numeric	MolWeight(g/mol)=633.6	InchiKey=OIZGSVFYBZVIK-FHHHURISA-N	
9	3SL_mol				[0, 0.01]	mol/L	Numeric			
10	6SL		SL-OS	6-Sialyl Lactose	[0, 1]	g/L	Numeric	MolWeight(g/mol)=633.5	InchiKey=MBSOBTIATUNICK-GIGDIUIZSA-N	
11	6SL_mol				[0, 0.01]	mol/L	Numeric			

Figure 15. The **Component** sheet

The **Component** sheet (Figure 15) has the heading (field names) as below:

Component | **Alternative** | **Group** | **Description** | **ValueSet** | **Unit** | **TYPE** | **MolWeight** | **ChemID** | **Comment**

- *Component*: The name of the component. It links an entry in the *Component* field of the **Master** sheet to its definition in the **Component** sheet. The first column must be unique; it is, generally an abbreviation of the chemical component. For instance: amino acids are named after the first 3 letters of their names added by an “A”, so Alanine, Arginine and Asparagine are called: “AlaA”, “ArgA” and “AspA”, respectively.
- *Alternative*: Other alternative(s) for the same component, separated by semi-colon (e.g.: AscorbicAcid for Vitamin C).
- *Group*: The group in the tree structure, which the component belongs to.

E.g.: Vitamin B1 (VitB1) belongs to vitamin B complex group (VitB), which belongs to the Vitamins group (Vit) which belongs to the Human Milk root (HM). **Figure 16** shows the tree structure illustrating the example mentioned before, where the leaves represent the component “VitB1”.

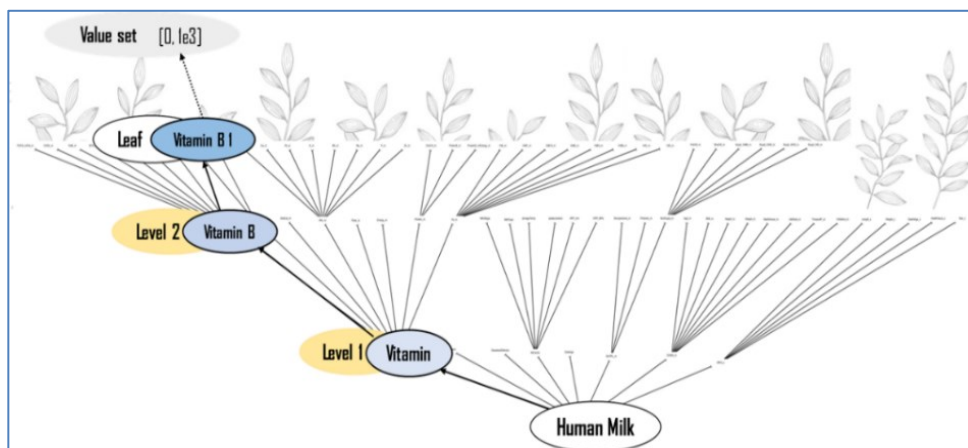


Figure 16. Hierarchical, tree-structure-like organisation of the value set of the Components, with Vitamin B1 a “leaf” of the tree.

- *Description*: A brief description of the component.
- *ValueSet*: The interval, from which the values for the component can be chosen, in the unit below (e.g. [0,10]).
- *Unit*: The physical unit of the above value. Allowed units are in the **Unit** sheet.
- *Type*: All are “Numeric” in the **Component** sheet
- *MolWeight*: Molecular weight of the component (if available) in g/mol.³ The molecular weight can be used to convert component values, given originally in Mol in the publication, to density in g/L.
- *ChemID*: Chemical ID of the component; most frequently InChiKey⁴, the IUPAC (International Union of Pure and Applied Chemistry) International Chemical Identifier⁵, which may be used for further search about this component.
- *Comment*: Free text about any more information.

InputBy sheet

Summary: The **InputBy** sheet shows the responsible for inputting data in *MilkyBase*

- *InputBy*: Short name for the responsible (e.g., JS for John Smith or UniDeb for the University of Debrecen)
- *Alternative*: Complete name.
- *Description*: A brief background of the responsible.
- *Comment*: Free text about any further information.

Unit sheet

Summary: This sheet contains the description of the units used in the database and their conversion rate.

	A	B	C	D	E	F	G
1	Unit	Alternative	Group	Description	Comment	FactorToBase	Basic_Unit
11	length		Unit				
12	m		length	meter	= 1.e3 mm	1000	mm
13	cm		length	centimeter	= 10 mm	10	mm
14	mm		length	milliMeter	basic unit of length	1	mm
15	mass		Unit				
16	kg		mass	kilogram	= 1000 gram	1.E+03	g
17	g		mass	gram	basic unit of mass	1	g
18	mg		mass	milliGram	=10 [^] (-3) gram	1.E-03	g
19	mcg		mass	microGram	=10 [^] (-6) gram	1.E-06	g
20	ng		mass	nanoGram	=1.e-9 gram	1.E-09	g

Figure 17. The Unit sheet

The Unit sheet (**Figure 17**) has the heading (field names):

Unit | *Alternative* | *Group* | *Description* | *Comment* | *FactorToBase* | *BasicUnit*

- *Unit*: Name of the unit as the *Component* or the *Condition* values is given. The basic units are in grey.
- *Alternative*: Other alternative names for the Unit (like Celsius instead of C).
- *Group*: Group where the unit belongs to.

The main groups (with their basic unit) are ratio (-), time (day), length (mm), mass (g), volume (L), dose (g/day), density (g/L). The latter one assumes that 1 L milk is 1 kg.

- *Description*: Brief description of the unit.
- *Comment*: Free text about any additional information.
- *FactorToBase*: Conversion rate to base unit.⁶
- *BasicUnit*: The basic unit of the group to which the unit belongs.

DynVal sheet

Summary: The values in the *Component* and *Condition* fields can be time-dependent series of numerical values, called dynamic values. These values are stored in a dedicated table called **DynVal** sheet. The pointer in the *ValueSet* of the *Component* or *Condition* fields of the **Master** sheet needs the initial “!” to link it to the table where the two columns of [time, value] are stored in the **DynVal** sheet. See **Figure 18**.

	A	B	C	D	E	F	G	
1	Key	Food	Source	Region	Cohort	MeasMe	Condition	Component
21	HM-MM-Apa-20-1	HumanMilk	Aparicio_20	Netherlands	51	ELISA; Mag	GestAge(week)=3	IL6(g/L)=!Aparicio_IL6 IgA(g/L)=!Aparicio_Ig

	A	B	C	D	E	F	G	H	I	J
1	DynVal	sampleSize	time	timeSD	timeLo	timeHi	timeUnit	Component	value	valueSD
139	Aparicio_IL6	28	14.75	1.84			day	IL6	9.36E-09	
140	Aparicio_IL6	16	43.58	5.02			day	IL6	5.32E-09	
141	Aparicio_IL6	11	85.35	2.33			day	IL6	2.64E-09	

Figure 18. Pointers start with exclamation mark and refer to a table with the same name (without the exclamation mark).

In the example above, the first column of the **DynVal** sheet identifies the dynamic response, as indicated by the initial “!” in the **Master** sheet. For example, ‘!Aparicio_IL6’, in the **Master** sheet, points to a temporal series of [time, value] pairs stored in the **DynVal** sheet with the first column entry “Aparicio_IL6”.

The **DynVal** sheet has the heading (field names):

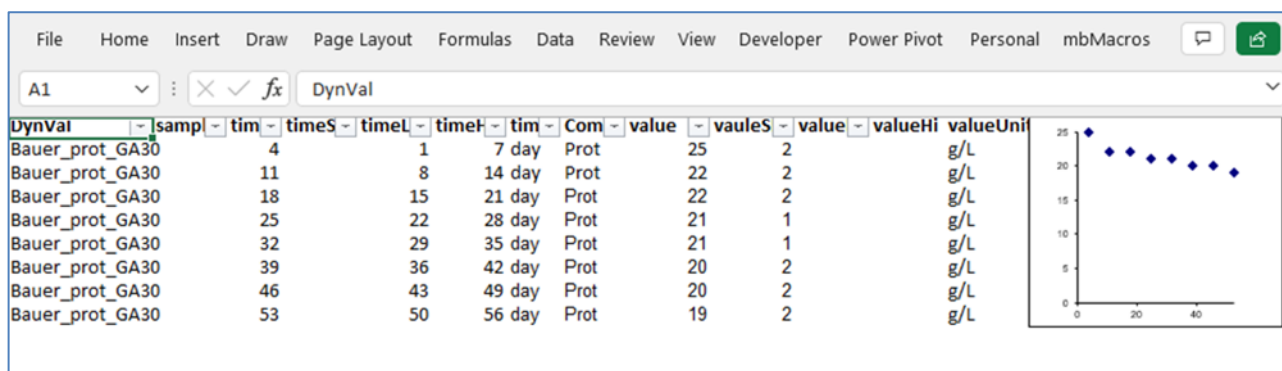
DynVal | sampleSize | time | timeSD | timeLo | timeHi | timeUnit | Component | value | valueSD | valueLo | valueHi | valueUnit | Comment

- *DynVal*: pointer variable (without “!”). It links the entries in the **Master** sheet to their definition in the **DynVal** sheet.
- *sampleSize*: It is possible that the sample size is not the same for all time points, this is why this field is introduced, though it is also featured in the **Master** sheet
- *time*: Sampling date in days related to birth (prePartum < 0; postPartum > 0). Type: Numeric; Default value set: [-1000,10000]
- *timeSD*: Standard deviation (SD) of the *time* Type: Numeric; Default value set: [-1000,10000]
- *timeLo*: A lower bound for the *time*. Type: Numeric; Default value set: [-1000,10000]
- *timeHi*: An upper bound for the *time*. Type: Numeric; Default value set: [-1000,10000]
- *timeUnit*: Time unit used (default: “day”).
- *value*: The actual quantity of the compound in the unit defined in the record. Type: Numeric; Default value set: [0,1e9]
- *valueSD*: Standard Deviation (SD) of the *value*. Type: Numeric; Default value set: [0,1e9]
- *valueLo*: A lower bound for the *value*. Type: Numeric; Default value set: [0,1e9]
- *valueHi*: An upper bound for the *value*. Type: Numeric; Default value set: [0,1e9]

- *valueUnit*: Unit in which the value is measured. Allowed units are in the **Unit** sheet.
- *Comment*: Free text about any specific information on the measurement.

Plot sheet

Summary: This sheet only serves for a storage area and the user is not supposed to change anything here. See **Figure 19**.



*Figure 19. The **Plot** sheet in a secondary window.*

Miscellaneous

Separators

The separators are used in this priority order:

- vertical bar (for vector-fields only, to separate vector-entries)
E.g. MilkStage(day)=3 | fishOil_m(g/day)=3.7
- semi-colon (for lists). If it is in a value set then it is convenient but not compulsory to enclose them by braces, e.g. {FM; HPLC; BM}
- comma (only for intervals in numeric fields as above). Separates the lower and upper limit of the interval
- point: Decimal symbol
- Allowed special characters in variable names: underscore, hyphen, plus-sign, and forward slash.
A convenient notation: X+Y is the name of a substance consisting of X and Y.
- X/Y is the name of the quantity which expresses the ratio between the respective amounts. So, for example, X/X+Y is quantifiable by the ratio in what proportion the X is part of the compound X+Y (note that the mathematical priority order between / and + is not valid here).
- Parentheses and dots are not supported in strings, except in the free text areas.
- Common names given to the fatty acids based on their structure (C:D).

Names for various Fatty Acids

See the reference:

- <https://pubmed.ncbi.nlm.nih.gov/>

Value conversion

- When values were g, mg etc. per 100 g edible food, they were converted to g/litre (or g/L) where 1 litre of milk is assumed to be 1kg.⁶

References

- 1 Krachler, M., Shi Li, F., Rossipal, E. & Irgolic, K. J. Changes in the Concentrations of Trace Elements in Human Milk during Lactation. *Journal of Trace Elements in Medicine and Biology* **12**, 159-176, doi:[https://doi.org/10.1016/S0946-672X\(98\)80005-9](https://doi.org/10.1016/S0946-672X(98)80005-9) (1998).
- 2 Ahmed, L., Islam, S. K. N., Khan, M. N. I., Huque, S. & Ahsan, M. Antioxidant Micronutrient Profile (Vitamin E, C, A, Copper, Zinc, Iron) of Colostrum: Association with Maternal Characteristics. *Journal of Tropical Pediatrics* **50**, 357-358, doi:10.1093/tropej/50.6.357 (2004).
- 3 NIH. *PubChem*, <<https://pubchem.ncbi.nlm.nih.gov/>>
- 4 Heller, S. R., McNaught, A., Pletnev, I., Stein, S. & Tchekhovskoi, D. InChI, the IUPAC International Chemical Identifier. *Journal of Cheminformatics* **7**, doi:10.1186/s13321-015-0068-4 (2015).
- 5 IUPAC. *THE IUPAC INTERNATIONAL CHEMICAL IDENTIFIER (INCHI)*, <<https://iupac.org/who-we-are/divisions/division-details/inchi/>> (
- 6 NIST. *Unit Conversion*, <<https://www.nist.gov/pml/weights-and-measures/metric-si/unit-conversion>>

Servicing macros to enter and collect data in MilkyBase

Once the MilkyBase.xlsx file is open, the user may also want to open the file “MBmacros.xlsm” containing “servicing macros”, which are useful to validate the entered data. They are written in Excel VBA and run in Excel 2016 or above. To use this file, it is necessary to enable the use of VBA macros in Excel. *It assumes that user works with Excel with comma for a separator and dot for the decimal point.*

Upon loading MBmacros.xlsm macros can be run from the Excel Macros menu.: See **Figure 20** .

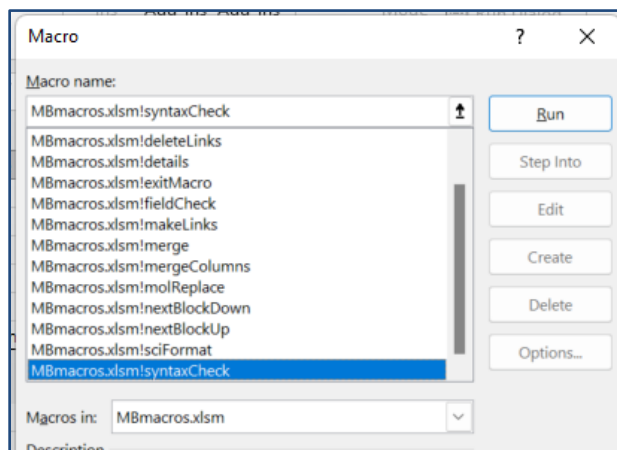


Figure 20. The MBmacros

Details references

Opens a window with the detailed explanation of the cell under the cursor. The shown information is either:

- details about the field to which the selected cell belongs
- details about the entry under the cursor. Generally, they are in the respective definition sheets. If the entry is a list, then the details are about the first item
- goes to the data where the user is referred to; possibly plot a temporal profile identified by a pointer.

All these appear in a secondary window. THE WINDOW SHOULD BE CLOSED after inspection

Syntax check (of selection)

Checks the validity of the selected cells (whether they satisfy the prescribed syntax of the field to which the cell belongs). You can check syntactically maximum 2000 rows in one run.

Separation into columns

The *Condition* and *Component* columns may contain items separated by vertical bars (|). All the cells in the selected region (**Figure 21**) will undergo this separation in a new sheet in such a way that the identical variables will be under each other (**Figure 22**), i.e., they will form a proper field. This step is necessary before using the “Details, references” macro for cells in the *Condition* and *Component* columns.

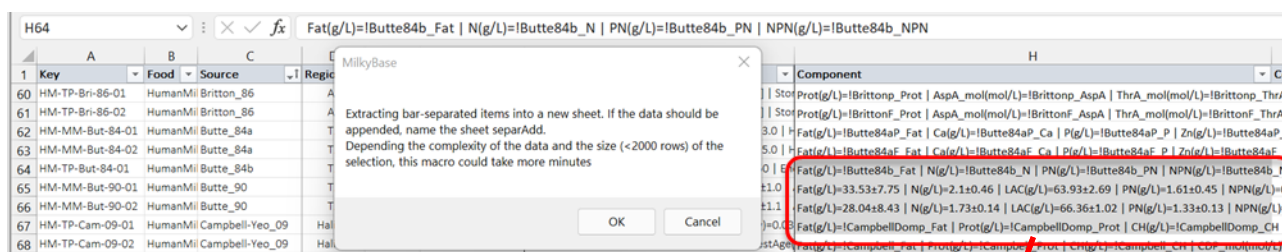


Figure 21. Separation into columns

key	Fat(g/L)	se.Fat(g/L)	lo.Fat(g/L)	hi.Fat(g/L)	N(g/L)	se.N(g/L)	lo.N(g/L)	hi.N(g/L)	PN(g/L)	se.PN(g/L)	lo.PN(g/L)	hi.PN(g/L)	NPN(g/L)	se.NPN(g/L)	lo.NPN(g/L)	hi.NPN(g/L)	LAC(g/L)	CH(g/L)	CDP_mol(mol/L)
HM-TP-But-84-01	IButte84b_Fat				IButte84b_N				IButte84b_PN				IButte84b_NPN						
HM-MM-But-90-01	33.53	7.75			2.1	0.46			1.61	0.45			0.48	0.1					
HM-MM-But-90-02	28.04	8.43			1.73	0.14			1.33	0.13			0.4	0.05					
HM-TP-Cam-09-01	ICampbellDomp_Fat																		

Figure 22. The separated columns

Merge columns

The inverse of the above operation; the selected region (**Figure 23**) will be put in one single cell, with vertical bars (|) separating the items which used to occupy an independent column (**Figure 24**).

key	Fat(g/L)	se.Fat(g/L)	lo.Fat(g/L)	hi.Fat(g/L)	N(g/L)	se.N(g/L)	lo.N(g/L)	hi.N(g/L)	PN(g/L)	se.PN(g/L)	lo.PN(g/L)	hi.PN(g/L)	NPN(g/L)	se.NPN(g/L)	lo.NPN(g/L)	hi.NPN(g/L)	LAC(g/L)	CH(g/L)	CDP_mol(mol/L)
HM-TP-But-84-01	IButte84b_Fat				IButte84b_N				IButte84b_PN				IButte84b_NPN						
HM-MM-But-90-01	33.53	7.75			2.1	0.46			1.61	0.45			0.48	0.1					
HM-MM-But-90-02	28.04	8.43			1.73	0.14			1.33	0.13			0.4	0.05					
HM-TP-Cam-09-01	ICampbellDomp_Fat																		

Figure 23. Merge columns

key	Fat(g/L)	se.Fat(g/L)	lo.Fat(g/L)	hi.Fat(g/L)	N(g/L)	se.N(g/L)	lo.N(g/L)	hi.N(g/L)	PN(g/L)	se.PN(g/L)	lo.PN(g/L)	hi.PN(g/L)	NPN(g/L)	se.NPN(g/L)	lo.NPN(g/L)	hi.NPN(g/L)	LAC(g/L)	CH(g/L)	CDP_mol(mol/L)
HM-TP-But-84-01	Fat(g/L)=IButte84b_Fat N(g/L)=IButte84b_N PN(g/L)=IButte84b_PN NPN(g/L)=IButte84b_NPN																		
HM-MM-But-90-01	Fat(g/L)=33.53±7.75 N(g/L)=2.1±0.46 PN(g/L)=1.61±0.45 NPN(g/L)=0.48±0.1 LAC(g/L)=63.93±2.69																		
HM-MM-But-90-02	Fat(g/L)=28.04±8.43 N(g/L)=1.73±0.14 PN(g/L)=1.33±0.13 NPN(g/L)=0.4±0.05 LAC(g/L)=66.36±1.02																		
HM-TP-Cam-09-01	Fat(g/L)=ICampbellDomp_Fat Prot(g/L)=ICampbellDomp_Prot CH(g/L)=ICampbellDomp_CH CDP_mol(mol/L)=ICampbellDomp_CDP Prolactin(g/L)=ICampbellDomp_Prolactin																		

Figure 24. The merged columns

Block downwards

The macro will select in the current column those cells which are under the current cell (where the cursor is) and their content is the same as under the cursor.

Block upwards

As above but selecting the part of the column OVER the current cell.

Scientific format

This macro examines all the values given to variables in the selected region and converts those which are less than 10^{-3} into scientific format.

Close

Closing “MBmacros.xlsm”.

The user may want to test the xlam version, too, available in the same FigShare folder. It has a user-friendly menu system; however that is not guaranteed to work on the various versions of MS Excel.